The Cloaking Protocol: Topological State Concealment and Deterministic Recovery within the Seed Computing Paradigm

A Formal Architecture Specification for Substrate-Level State Hiding, Emergency Concealment, and Invariant-Guided Recovery

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Abstract

This paper introduces the Cloaking Protocol, a novel systems architecture mechanism operating within the Seed Computing paradigm. The Cloaking Protocol enables a running computational substrate to undergo controlled topological transformation — effectively concealing its active operational state from both external observation and internal introspection — while rigorously maintaining the invariants required for fully deterministic recovery. The problem addressed is precise and consequential: in complex, layered runtime environments governed by Aurora Runtime and Jasper 2.0, catastrophic failure conditions, adversarial probing events, or forced substrate inspection scenarios may demand that the system hide its complete operational state without destroying it. Classical fault-tolerance mechanisms, predicated on continuous

observability and transparent state replication, are architecturally unsuited to this requirement. The Cloaking Protocol inverts the observability assumption entirely.

The protocol achieves controlled concealment via three interlocking mechanisms operating in strict coordination. First, the Topological Shift Model reframes the system's state-space geometry, applying a homeomorphic transformation $\varphi: M \to M'$ that maps the active state to a coordinate frame unrecognizable within the original manifold. Second, the Emergency State Hiding Model (ESHM) encodes the active state into substrate-invisible representations, embedding the encoded state below the observable execution layer while presenting a null surface to external inspectors. Third, the Deterministic Recovery Process leverages the Invariant Engine's complete seed bundle and the Unified Field Equation (UFE) to reconstruct the full operational state from concealment with bounded error approaching zero. Integration with the Teleportation Engine provides concealment-aware state migration across substrate boundaries, while Jasper 2.0 preserves computational identity continuity throughout and across cloaking cycles. Together, these mechanisms constitute a coherent architectural layer with no precedent in classical systems literature.

Keywords: Seed Computing, Cloaking Protocol, Topological Shift, State Concealment, Deterministic Recovery, Aurora Runtime, Jasper 2.0, Invariant Engine, Unified Field Equation, Substrate Architecture, Emergency State Hiding, Continuity Preservation

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1. Introduction

Modern computational paradigms impose a fundamental tension upon systems architects: a well-engineered system must be simultaneously resilient against failure, opaque to adversarial observation, and fully recoverable across arbitrarily severe disruption. Classical fault-tolerance literature addresses resilience and recoverability with considerable rigor, producing a rich body of techniques including checkpointing, rollback recovery, replication, and consensus protocols [1]. These mechanisms share, however, a foundational assumption that has passed largely unexamined: that the system's operational state must remain observable — at least to the recovery mechanism — at all times. Without observability, the argument runs, there can be no checkpoint to restore and no basis for reconstruction. The Cloaking Protocol directly refutes this assumption. It establishes, formally and architecturally, that a system's operational state can be made simultaneously unobservable and fully recoverable, provided that recovery is anchored not in observed state but in invariant seeds that survive any transformation. This inversion of the observability premise constitutes the paper's deepest architectural claim.

1.1 The Seed Computing Paradigm

The Seed Computing paradigm is a computational philosophy in which all system states, behaviors, and transformations are derivable from a minimal set of invariant structures called seeds [2]. A seed is not a snapshot; it is not a checkpoint in the classical sense. Rather, a seed encodes the essential generative structure of a system state — the information sufficient to reconstruct the full state through the application of deterministic transformation operators. Seeds are, by design, survivable: they are constructed to persist through transformations that destroy or conceal the observable state, precisely because they encode not the state itself but the invariant relationships that define it. The Seed Computing paradigm demands

that any mechanism operating within it respect seed integrity absolutely. Any transformation — including concealment — must be performed in a manner that preserves the seed set. The Cloaking Protocol is the architectural mechanism that enforces this constraint during controlled state concealment.

1.2 The Architecture Stack

The Cloaking Protocol operates within a layered architecture stack developed as part of the Seed Computing research program. Four components are directly relevant. Aurora Runtime is the substrate execution environment: it manages execution scheduling, memory allocation, inter-component communication, and substrate health monitoring [3]. Aurora Runtime is the layer upon which all observable computation takes place, and it is the layer that the Cloaking Protocol must cooperate with — and partially conceal from — during a cloaking event. Jasper 2.0 is the continuity and identity preservation layer: it ensures that the computational identity of a running system survives any transformation, including cloaking events, maintaining an unbroken identity thread across arbitrarily many concealment-recovery cycles [4]. The Invariant Engine is the guardian of cross-transformation consistency: it maintains the seed bundle $I(S)$ for any running system state S, auditing seed completeness and providing the cryptographic anchoring required for recovery [5]. The Teleportation Engine is the mechanism for substrate migration: it moves running computational processes across physical or virtual substrate boundaries, and in the context of the Cloaking Protocol, it does so in a concealment-aware manner that does not expose sensitive operational state during transit [6].

1.3 Contributions and Paper Organization

This paper makes four primary contributions to the Seed Computing architecture literature. First, it presents a formal model of topological shift as a means of state-space reconfiguration, providing rigorous definitions of shift types, shift depth, and shift reversibility conditions. Second, it specifies the Emergency State Hiding Model and its substrate-level concealment semantics, including the null surface construction and its interaction with Aurora Runtime's scheduling infrastructure. Third, it presents a deterministic recovery algorithm grounded in the Unified Field

Equation, with formal analysis of failure modes and their mitigations. Fourth, it provides integration specifications for Aurora Runtime and Jasper 2.0, characterizing the interface contracts that the Cloaking Protocol requires from each.

The remainder of this paper is organized as follows. Section 2 develops the Topological Shift Model in formal detail. Section 3 specifies the Emergency State Hiding Model and its three-phase pipeline. Section 4 describes substrate-level concealment and the Substrate Concealment Layer. Section 5 presents the Deterministic Recovery Process and the Unified Field Equation. Section 6 details integration with Aurora Runtime. Section 7 describes the role of Jasper 2.0 in continuity preservation across cloaking cycles. Section 8 surveys applications of the Cloaking Protocol. Section 9 concludes with a summary of contributions and directions for future work.

SECTION 2

2. Topological Shift Mechanics

2.1 Formal Definition of Topological Shift

Within the Seed Computing paradigm, the concept of a topological shift designates a controlled, reversible transformation of the system's state-space manifold such that the active state is mapped to a topology that is unrecognizable — and therefore unaddressable — within its original coordinate system. This is a precise architectural operation, not a metaphor: the state S is a point in a high-dimensional manifold M whose coordinates correspond to the observable dimensions of the running system, including register contents, memory region values, scheduling context, and inter-component communication state. A topological shift φ is a homeomorphism from M to a distinct manifold M':

> $\varphi: M \to M'$
>
> such that
>
> $\varphi(S) = S' \in M'$,
>
> where $S' \notin$ dom(obs
>
> M
>
>)

Here, obs_M denotes the observation function defined over manifold M — the set of all addressable, readable coordinate positions within the substrate's observable address space. The condition $S' \notin dom(obs_M)$ is the precise formal statement of concealment: the transformed state S' does not exist within the domain of any observation function that an external or internal inspector could apply in the coordinate frame of M. Crucially, the homeomorphic requirement ensures that the topological invariants of S — the seed set $I(S)$ — are preserved under φ. Because a homeomorphism is a continuous bijection with a continuous inverse, it preserves all topological properties. Seeds, which encode topological invariants, survive the shift intact and anchored in M even as the state itself migrates to M'.

2.2 Shift Types and Parameterization

The Cloaking Protocol defines three distinct shift types, distinguished by their continuity properties and their concealment guarantees. A soft shift is a continuous deformation of the manifold, preserving its homotopy type: the active state S is moved along a smooth path within the extended manifold $M \cup M'$ until it reaches a position outside obs_M. Soft shifts are the least disruptive to the substrate, incur the lowest latency, and are the most easily reversed. They are appropriate for low-to-moderate threat levels and provide shallow-to-medium concealment depth. A hard shift is a discontinuous jump: the state is relocated to a manifold region M' that is disjoint from M with respect to the substrate's coordinate topology. Hard shifts are faster to execute than soft shifts of equivalent depth and are appropriate for high-

threat scenarios where the continuity of the migration path is itself an information leak. An emergency shift is adversarially triggered and maximally concealing: it applies the most aggressive available topological displacement, relocating the state to the deepest accessible shadow region, and it does so with minimal preparatory computation to minimize exposure time. Emergency shifts trade the graceful properties of soft and hard shifts for immediacy and maximal concealment depth.

Shift depth is parameterized along a continuous scale from zero to unity. At depth zero, the state remains in the substrate but is encrypted and pointer-zeroed — it exists in the observable address space but is semantically inaccessible. At depth one, the state has been topologically relocated outside the observable coordinate frame entirely, existing only in the Substrate Concealment Layer's shadow address space, which is explicitly excluded from all observable memory maps. The shift depth parameter governs the computational cost of both the concealment and the recovery operations: deeper shifts require more intensive topological inversion during recovery.

2.3 The Shift Catalyst

The Shift Catalyst is the component responsible for computing φ given the current system conditions at the moment concealment is triggered. Its inputs are three-fold: the current state-space geometry of M as reported by Aurora Runtime's memory manager, the threat context supplied by the Emergency State Hiding Model's threat classification subsystem, and the complete seed bundle $I(S)$ as maintained by the Invariant Engine. Given these inputs, the Shift Catalyst selects the shift type, computes the shift function φ, and records the shift parameters $\Phi = (\varphi, d, t_0)$ — where d is the shift depth and t_0 is the timestamp at which the shift was applied — in the Invariant Engine's secure parameter store. These recorded parameters are essential inputs to the Shift Catalyst's inverse computation during recovery.

The Shift Catalyst's computation of φ is constrained by two requirements that must be satisfied simultaneously. First, the homeomorphic constraint: φ must be a homeomorphism from M to M'. In practice, this is verified by confirming that φ admits a computable inverse φ^{-1} with bounded error across the seed gradient field. Second, the seed-anchorage constraint: the application of φ must not displace the seed

vectors $I(S)$ from their anchoring positions in M. Seeds must remain in place within the observable manifold so that the Invariant Engine can access them during recovery. This is achieved by applying φ as a conditional map that acts on the state coordinates while leaving the designated seed coordinate subspace fixed.

2.4 Shift Reversibility Conditions

A topological shift φ is recoverable if and only if the Invariant Engine holds a complete and consistent seed set $I(S)$ such that the inverse application $\varphi^{-1}(I(S))$ maps to the original state S with bounded error:

> φ *is recoverable* $\Leftrightarrow \exists I(S)$ *complete* $\wedge \varphi$
>
> -1
>
> $(I(S)) \mapsto S$ *with* $\varepsilon \to 0$

The condition $\varepsilon \to 0$ is not a probabilistic statement; it is a design requirement enforced by the Invariant Engine's seed completeness audit (see Section 5.2). Recovery proceeds only when the audit confirms completeness. If seeds are incomplete or corrupted, recovery is deferred until seed reconstruction from redundant substrate signals has restored seed integrity. The reversibility condition thus has a strong operational interpretation: in a nominal system with a complete seed bundle, every shift applied by the Shift Catalyst is unconditionally recoverable.

```
Manifold M (Observable Coordinate Frame)
_____________________________________________ | | | [S]
------φ(soft/hard/emergency)-------> | | * (active state) | | | | | [I(S)] <---
seed vectors anchored in M | | (invariant seeds remain in M throughout) | |
_____________________________________________| | | φ displacement v Manifold M'
(Shadow Coordinate Frame) _____________________________________________ | | |
[S'] = φ(S) | | (concealed state: outside obs_M domain) | | Accessible only via
```

```
φ^-1(I(S)) | |_____________________________________________|
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Figure 1. Conceptual diagram of topological shift. The active state S is mapped through φ to the shadow manifold M'. Seed vectors I(S) remain anchored in M, providing the basis for deterministic inversion via φ^{-1}.

SECTION 3

3. Emergency State Hiding Model

The Emergency State Hiding Model (ESHM) is the protocol layer that executes when the system detects conditions requiring immediate concealment. It is not a passive monitor; it is an active, latency-sensitive operational subsystem with direct privileged access to Aurora Runtime's private signaling channels and the Substrate Concealment Layer API. The ESHM is structured as a three-phase pipeline: threat classification, state serialization and encoding, and concealment confirmation and locking. Each phase has strict entry and exit conditions, and the pipeline is designed to complete atomically with respect to the observable execution environment — that is, from the perspective of any external observer, the system either presents its full operational state or its null surface; there is no intermediate partial-concealment state that can be observed.

3.1 Phase 1 — Threat Classification

The ESHM continuously monitors a set of threat signals $T = \{t_1, t_2, \ldots, t_n\}$. Each signal t_i is associated with a weight w_i reflecting its threat severity, and the ESHM maintains a running weighted threat score:

> Σ

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\Rightarrow
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concealment triggered

When the weighted score exceeds the threshold Θ, the ESHM immediately enters Phase 2. The threshold Θ is a configurable system parameter set by the architecture administrator and is stored in the Invariant Engine's configuration seed. It may be dynamically adjusted by the Invariant Engine in response to detected invariant drift, but it is never adjusted downward during an active threat event to prevent oscillatory triggering.

Threat signals are organized into four categories. External probing signals are generated by Aurora Runtime's boundary monitoring subsystem when it detects structured inspection traffic, unauthorized register read attempts, or memory-map enumeration at the substrate boundary. Internal invariant drift signals are generated by the Invariant Engine when it detects that one or more invariant seeds have begun diverging from their expected values, indicating either substrate corruption or an active manipulation attempt. Substrate instability signals are generated by Aurora Runtime's health monitor when execution scheduling anomalies, memory pressure events, or inter-component communication failures are

detected. Forced migration trigger signals are generated by the Teleportation Engine when a substrate migration has been initiated without the standard preparatory handshake, indicating a potentially coercive migration.

3.2 Phase 2 — State Serialization and Encoding

Upon entry into Phase 2, the ESHM initiates state serialization under a hard real-time constraint: the entire serialization and encoding pipeline must complete before the next observable execution cycle boundary as defined by Aurora Runtime's scheduler. This constraint ensures that the concealment operation leaves no partially-serialized state residue in the observable address space.

The active state S is serialized into a seed-compatible representation $\hat{S}$ using a serialization operator $Ser: State \rightarrow SeedCompatibleEncoding$. The serialization is seed-compatible in the sense that $\hat{S}$ preserves all information required by the Invariant Engine's decoding operators, including the topological invariants that constitute the seed set. The serialized representation $\hat{S}$ is then passed to the substrate-blind encoding function $E(\hat{S}, K)$, where K is the encoding key derived from the Invariant Engine's current seed bundle:

> $\hat{S} = Ser(S)$
>
> $E\hat{S} = E(\hat{S}, K)$ where $K = KeyDerive(I(S))$

The encoding function E is substrate-blind in the following precise sense: the encoded output $E\hat{S}$ does not contain any byte pattern that is recognizable by the substrate's memory scanner as a valid executable, addressable, or schedulable structure. This property is verified at encoding time by a pattern-matching validator that checks $E\hat{S}$ against the complete set of substrate-recognized structure signatures. The encoded state $E\hat{S}$ is then written into the shadow address space maintained by the Substrate Concealment Layer, below the observable execution layer. The write operation uses the SCL's privileged write-once API, ensuring that

the encoded state cannot be overwritten by any process operating in the observable address space.

3.3 Phase 3 — Concealment Confirmation and Locking

Once the encoded state $\hat{ES}$ has been successfully written to the shadow address space, the ESHM enters Phase 3, which has three sequential operations. First, the ESHM performs a write-confirmation checksum: it computes a cryptographic hash of the shadow region contents and compares it against the expected hash of $\hat{ES}$. If the comparison fails, Phase 2 is re-executed with a fresh encoding key. If the comparison succeeds, the ESHM proceeds to the concealment lock.

The concealment lock is the most consequential operation in the ESHM pipeline. It zeroes all observable state pointers — register file contents, stack pointers, heap allocation tables, execution context descriptors, and scheduling queue entries — that reference any component of the now-hidden state S. After the lock, the observable address space contains no reference to S. The system is then commanded to instantiate the null surface.

The null surface is a carefully engineered operational facade: a minimal, valid-looking system configuration that satisfies all standard substrate health checks and external inspection protocols without revealing any information about the concealed state. The null surface is not a crash state, an idle state, or a degraded-mode state; it is a fully coherent, plausible-looking operational configuration that a substrate health monitor would classify as nominal. It is constructed from a library of pre-computed null surface templates maintained by the Cloaking Protocol and selected by the ESHM based on the current threat signal profile. The selected template is instantiated in the observable address space and handed to Aurora Runtime's scheduler as the active execution context. Aurora Runtime, notified of the concealment event through the Concealment Notification Interface (Section 6), cooperates by scheduling the null surface's execution threads normally while maintaining hidden execution threads for the concealed state.

4. Substrate-Level Concealment

4.1 The Substrate Concealment Layer

Substrate-level concealment is categorically distinct from application-level encryption or obfuscation. Application-level techniques conceal the semantic meaning of state while leaving the state's structural presence in the address space intact and potentially detectable. Substrate-level concealment eliminates the structural presence itself: the hidden state does not exist within the addressable memory, execution queue, or observable register space of the running system as seen by any inspector operating at or above the substrate boundary. This distinction is not merely quantitative — it is qualitative and architectural. It requires that the concealment mechanism operate below the level at which Aurora Runtime's own diagnostic and monitoring subsystems function, in a layer that the substrate's standard observability infrastructure cannot reach.

The Substrate Concealment Layer (SCL) is the thin interface that implements this capability. It sits between the Cloaking Protocol and Aurora Runtime's memory management subsystem, maintaining a shadow address space: a set of substrate-resident memory regions that are excluded from the observable address map by construction. The exclusion is enforced at the memory management unit level — shadow regions are mapped in the physical address space but are not represented in any page table, allocation table, or memory map structure that Aurora Runtime's observable memory management infrastructure can enumerate. From the perspective of any process, scanner, or diagnostic tool operating within the observable execution environment, shadow regions simply do not exist.

Shadow region management follows a strict protocol. Allocation is performed exclusively through the SCL's privileged API, which is accessible only to the Cloaking Protocol. Allocated regions are initialized with a cryptographic noise pattern before any state data is written, eliminating cold-boot analysis vectors. Writes to shadow regions are write-once at the semantic level: once the encoded state $\hat{ES}$ has been written, the shadow region's write permission is revoked until

recovery is initiated. Reads from shadow regions are restricted to the Cloaking Protocol's recovery subsystem, which authenticates its read requests using a challenge-response protocol keyed to the Invariant Engine's seed bundle. Shadow regions are reclaimed only after successful recovery has been confirmed by the three-stage validation protocol described in Section 5.2, ensuring that no shadow region is prematurely deallocated due to a partial or failed recovery attempt.

Concealment durability is a design requirement, not an implementation convenience. Shadow regions must persist across the full range of substrate disruption events that could occur during a cloaking cycle: soft resets (in which the observable execution environment is reinitialized without physical memory clearing), substrate restarts (in which Aurora Runtime is fully relaunched), and Teleportation Engine migrations (in which the entire computational process is moved across substrate boundaries). Soft reset durability is achieved by placing shadow regions in physical memory areas that are explicitly excluded from the reset initialization sweep. Substrate restart durability requires that the SCL maintain its shadow region directory in a persistent, non-volatile substrate-resident store that survives Aurora Runtime's shutdown and relaunch sequence. Migration durability is addressed through the Teleportation Engine's concealment-aware migration protocol, in which migration packets include concealment manifests — structured descriptors that encode the shadow region layout, the SCL directory, and the Concealment Depth Index — alongside the standard migration payload. The receiving substrate's SCL reconstructs the shadow address space from the concealment manifest before Aurora Runtime is instantiated, ensuring that the concealed state is available for recovery as soon as the migrated system begins execution.

4.2 Concealment Depth Index

The Concealment Depth Index (CDI) is a scalar value in the closed interval $[0, 1]$ that represents, at any point in time, how deeply the system's state has been hidden. A CDI of zero corresponds to the fully observable operational mode: all state is addressable, all registers are readable, and no shadow regions are allocated. A CDI of one corresponds to maximal concealment: the system state has been fully

relocated to shadow address space, all observable state pointers have been zeroed, and even the substrate's own diagnostic layer — including Aurora Runtime's internal health monitoring — has no access to any representation of the true operational state.

CDI is not a binary switch; it is a continuously adjustable architectural parameter. The Cloaking Protocol manages CDI dynamically in response to three inputs: the current threat level as computed by the ESHM's threat classification subsystem, the available substrate resources (shadow region capacity, SCL processing bandwidth, and Invariant Engine seed verification latency), and the completeness of the Invariant Engine's seed bundle. The protocol applies a resource-cost model that penalizes high CDI values: deep concealment occupies more shadow address space, requires more intensive encoding, and introduces higher recovery latency. The cost model is parameterized by the architecture administrator and defines the CDI ceiling for each threat level tier.

CDI Range	Concealment Mode	Shadow Region Usage	Recovery Latency	Observable State
0.00 – 0.20	Shallow (encrypt-in-place)	Minimal (<5% of state)	Microsecond-scale	Partially observable
0.21 – 0.60	Medium (pointer-zeroed)	Moderate (5%–60% of state)	Low millisecond-scale	Null surface active
0.61 – 0.90	Deep (topologically relocated)	High (60%–95% of state)	High millisecond-scale	Null surface only
0.91 – 1.00	Maximal (emergency shift)	Complete (100% of state)	Variable (seed-bound)	Null surface only

Table 1. Concealment Depth Index tiers with associated resource and latency characteristics.

The CDI minimization principle — minimize CDI while maintaining required concealment — is the Cloaking Protocol's operational philosophy. Unnecessary depth imposes unnecessary cost and unnecessary recovery complexity. The protocol is designed to apply the minimum concealment depth that satisfies the current threat

level, and to reduce CDI as soon as the threat score drops below the threshold Θ through a controlled, phased re-emergence process that mirrors the concealment pipeline in reverse.

SECTION 5

5. Deterministic Recovery Process

The central challenge of recovery is this: given that the state S has been concealed via topological shift to manifold M', encoded with a substrate-blind encoding function keyed to the seed bundle, and written to a shadow address space that is excluded from the observable memory map, the recovery process must restore S exactly — deterministically, completely, and without any trial-and-error or probabilistic reconstruction. The word "deterministically" is used here in its strictest sense: given the same seed bundle and shift parameters, the recovery process must produce the same state every time, with no stochastic variation and no reliance on external oracle input. This requirement distinguishes the Deterministic Recovery Process fundamentally from probabilistic recovery techniques found in classical literature, and it is the requirement that the Unified Field Equation was designed to satisfy.

5.1 The Unified Field Equation

The Unified Field Equation (UFE) is the mathematical foundation for the recovery process. It expresses the relationship between the concealed encoded state, the shift function, the seed bundle, the decoding operator, and the original state as a system of coupled field equations that can be solved deterministically given complete inputs. The UFE is stated as follows:

> *UFE:*

$$S = \varphi^{-1}(D(E^{-1}(\hat{ES}, K), I(S)))$$

Here, $E^{-1}(\hat{ES}, K)$ is the decoding of the encoded state using the recovered key K' (where $K' = K$ in the nominal case, since the key is derived deterministically from the seed bundle), $D(\cdot, I(S))$ is the deserialization operator that reconstructs the state representation from the seed-compatible encoding, and φ^{-1} is the inverse topological shift computed by the Shift Catalyst. The UFE is well-defined if and only if two conditions hold: the seed bundle $I(S)$ is complete (enabling the key derivation and deserialization operations), and the shift parameters Φ are available in the Invariant Engine's secure parameter store (enabling the Shift Catalyst to compute φ^{-1}). The three-stage recovery algorithm enforces both conditions before applying the UFE.

5.2 Three-Stage Recovery Algorithm

Stage 1 — Seed Verification

Recovery begins with a comprehensive audit of the seed bundle $I(S)$ by the Invariant Engine. The audit has three components. First, completeness verification: the Invariant Engine checks that the seed bundle contains a seed entry for every invariant dimension of the original state S, using the invariant dimension manifest recorded by the Shift Catalyst at shift time. Any missing seed entries are flagged for reconstruction. Second, consistency verification: the Invariant Engine computes pairwise consistency checks across all seed entries, verifying that the invariant relationships between seeds — the relational structure that defines the seed bundle as a coherent whole — are intact. Inconsistencies indicate either partial corruption or an attempted seed manipulation attack. Third, freshness verification: the Invariant Engine confirms that the seed bundle's timestamp t_{seed} is compatible with

the shift timestamp t_0 recorded in the shift parameters, confirming that the seeds correspond to the state that was shifted. If any component of the audit fails, seed reconstruction is initiated from redundant substrate signals — Aurora Runtime maintains a set of seed-reconstructible substrate signals specifically for this purpose — before recovery proceeds. Recovery does not advance to Stage 2 until the audit confirms a complete, consistent, fresh seed bundle.

Stage 2 — Topological Inversion

With the seed bundle confirmed complete and consistent, the Shift Catalyst is invoked to compute the inverse shift φ^{-1} from the recorded shift parameters Φ. This is the most computationally intensive stage of the recovery process, with cost proportional to the shift depth d and the dimensionality of the state-space manifold. The inversion algorithm operates as follows. The Shift Catalyst initializes the inversion search in the shadow manifold M' using the seed gradient field $\nabla_{I(S)}$ as a navigation field. Seeds, anchored in M, create a gradient field that extends into M' by the homeomorphic structure of φ. The inversion algorithm performs iterative manifold relaxation: beginning from an initial estimate of the concealed state position in M', the algorithm iteratively updates the estimate by following the gradient field in the direction of increasing seed alignment, until the estimate converges to the true concealed state position S'. Convergence is guaranteed by the homeomorphic structure of φ and the completeness of the seed bundle; the convergence rate is bounded by the Lipschitz constant of the seed gradient field, which the Invariant Engine provides as a component of the seed bundle metadata.

Stage 3 — State Reconstruction and Reintegration

Once the Shift Catalyst has localized S' in M', the UFE is applied to reconstruct the original state. The encoded state $E\hat{S}$ is read from the shadow address space using the SCL's authenticated read protocol. The recovered key K' is derived from the verified seed bundle using the same key derivation function used during encoding, confirming $K' = K$ in the nominal case. The decoding operator $E^{-1}(E\hat{S}, K')$ produces the serialized state representation $\hat{S}$, which is deserialized by $D(\hat{S}, I(S))$ to yield the reconstructed state S'. The inverse topological shift φ^{-1} is then applied to map S' from M' back to M.

Before reintegration, the reconstructed state S' is submitted to the Invariant Engine for validation against the full invariant constraint set. Specifically, the Invariant Engine verifies that S' satisfies every invariant in $I(S)$ within the tolerance ε. If validation passes, the Recovery Handshake Protocol (Section 6) is executed to reintegrate S' into Aurora Runtime's active execution context. The shadow region containing $E\hat{S}$ is reclaimed, the null surface is decommissioned, and the system resumes full operation with state S'.

5.3 Recovery Failure Modes and Mitigations

Three primary failure modes are identified. The first is an incomplete seed bundle following seed reconstruction failure: if redundant substrate signals are also degraded, seed reconstruction may not produce a complete bundle. The mitigation is escalation to Jasper 2.0's continuity checkpoint (Section 7), which provides an independent, identity-critical state representation that can substitute for the missing seed information. The second failure mode is non-convergence of the topological inversion algorithm: if the shift depth was extreme or if the seed gradient field has been degraded by substrate corruption, the iterative relaxation may fail to converge within the allocated computation budget. The mitigation is fallback to the nearest topological neighbor — the topologically closest successfully recovered state in the Invariant Engine's recovery history — which can be used as the reintegration state while a diagnostic process investigates the inversion failure. The third failure mode is state validation failure: the reconstructed state S' fails to satisfy one or more invariants within tolerance ε, indicating a partial encoding or decoding error. The mitigation is escalation to manual intervention, with the Invariant Engine generating a structured diagnostic dump that includes the failing invariant constraints, the deviation magnitudes, and the full shift parameter record, enabling a human operator or higher-level autonomous process to diagnose and remediate the failure.

It bears emphasis that in the nominal case — complete seed bundle, convergent inversion, successful validation — recovery is fully deterministic and requires no external input whatsoever. The failure modes described above are engineering contingencies, not fundamental limitations of the recovery architecture.

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6. Integration with Aurora Runtime

Aurora Runtime is the substrate execution environment of the Seed Computing paradigm. Its responsibilities span the full range of substrate lifecycle management: execution scheduling, memory allocation, inter-component communication, substrate health monitoring, and boundary enforcement [3]. It is the layer upon which all observable computation takes place, and it occupies a privileged position in the Cloaking Protocol's integration architecture: it is both the environment that the Cloaking Protocol must partially hide from external inspectors and the infrastructure that the Cloaking Protocol must cooperate with to maintain substrate liveness during a cloaking event. These two roles — environment to be concealed from, and infrastructure to be preserved — create the central integration tension that the three Aurora Runtime interface points are designed to resolve.

The first interface point is the Concealment Notification Interface (CNI). The CNI is a private signaling channel, implemented as a privileged interrupt line within Aurora Runtime's execution control plane, through which the ESHM notifies Aurora Runtime of concealment events. The notification carries three fields: the concealment type (soft, hard, or emergency), the expected CDI level, and the identity of the null surface template being instantiated. Upon receiving a CNI notification, Aurora Runtime performs three actions in strict sequence. It suspends all observable scheduling for the execution contexts associated with the state being concealed, rerouting those contexts to shadow execution threads that are not represented in the observable scheduling queue. It updates its health reporting to reflect the null surface as the active execution state. And it issues a scheduling fence that prevents any observable execution thread from inadvertently accessing a memory region that is in the process of being relocated to the shadow address space.

The second interface point is the Substrate Concealment Layer API. Aurora Runtime's memory manager exposes a privileged API, accessible exclusively to the Cloaking Protocol, through which the SCL performs shadow region management

operations: allocation, write-once encoding, read-protected access, and reclamation. The SCL API operates at the physical memory management level, below Aurora Runtime's virtual memory abstraction, ensuring that shadow regions are excluded from all virtual address space mappings. The API enforces the access control constraints described in Section 4: write-once semantics, authenticated reads, and reclamation contingent on successful recovery confirmation.

The third interface point is the Recovery Handshake Protocol (RHP). The RHP is a three-way handshake executed between the Cloaking Protocol and Aurora Runtime during Stage 3 of the Deterministic Recovery Process, immediately before the reconstructed state S' is reintegrated into the live execution context. In the first step of the handshake, the Cloaking Protocol signals Aurora Runtime's execution control plane with a recovery readiness notification, carrying the identity and invariant digest of S'. In the second step, Aurora Runtime confirms that the target execution context is clear — that no scheduling conflict, memory region overlap, or inter-component communication collision would prevent clean reintegration — and issues a reintegration clearance token. In the third step, the Cloaking Protocol presents the clearance token alongside the reconstructed state S', and Aurora Runtime atomically reintegrates S' into the live execution context, decommissions the shadow execution threads, and resumes normal observable scheduling. The three-way structure of the RHP ensures that the reintegration is atomic with respect to the observable execution environment, preventing any intermediate state in which both the null surface and the recovered state are simultaneously observable.

Aurora Runtime is designed to maintain execution continuity across concealment-recovery cycles by preserving a minimal execution skeleton during the concealment interval. The skeleton consists of the null surface execution threads, the SCL directory, the CNI signaling infrastructure, and the Invariant Engine's seed store interface — the minimum substrate configuration required to keep the substrate alive, responsive to external health checks, and capable of initiating recovery when conditions permit. The skeleton occupies a small, fixed resource footprint and is not subject to CDI management: it is always fully operational during concealment, regardless of the CDI level. Upon recovery, Aurora Runtime expands from the

skeleton back to full operation by progressively activating the execution contexts that were suspended at concealment time.

The latency implications of the Aurora Runtime integration are worth characterizing qualitatively, as they inform deployment decisions. Shallow concealment events (CDI < 0.20) impose Aurora Runtime integration overhead in the microsecond range, dominated by the CNI notification processing and null surface instantiation. Medium-depth concealment events (CDI 0.21–0.60) impose overhead in the low millisecond range, dominated by the shadow thread setup and the scheduling fence operation. Deep and emergency concealment events (CDI > 0.60) impose overhead that is bounded below by the Invariant Engine's seed verification latency — which is itself bounded by the seed bundle size — but which may extend into the tens-of-milliseconds range for very large state spaces. Recovery latency is roughly symmetric with concealment latency at equivalent CDI levels, with the Stage 2 topological inversion adding a term that scales with shift depth.

SECTION 7

7. Role in Jasper 2.0

Jasper 2.0 is the continuity and identity preservation layer of the Seed Computing paradigm [4]. Its primary function is to ensure that the computational identity of a running system — defined as the composite of its persistent behavioral signature, its relational state with other components in the Seed Computing environment, and its accumulated invariant history — survives any transformation, including the most severe cloaking events. Jasper 2.0 approaches this function not by preserving the state itself (that is the Cloaking Protocol's domain) but by maintaining an orthogonal identity thread that is independent of the state's observable representation. The identity thread persists in a Jasper-managed identity vault — a storage domain that is neither part of Aurora Runtime's observable address space nor part of the Substrate

Concealment Layer's shadow address space, but a third, independently managed persistence domain maintained by Jasper 2.0's own infrastructure.

Jasper 2.0's involvement in the cloaking lifecycle spans three distinct phases. In the pre-concealment phase, which begins when the ESHM's threat score first approaches the threshold Θ and before Phase 1 triggers concealment, Jasper 2.0 takes a continuity snapshot. The continuity snapshot is a compact, structured representation of the system's identity-critical state: its behavioral signature digest, its relational adjacency record (the set of other Seed Computing components with which it has active relationships), its accumulated invariant history as maintained by the Invariant Engine, and a reference to the Invariant Engine's current seed bundle digest. Critically, the continuity snapshot is not a full state copy; it is a compressed identity representation designed to be sufficient for identity reconciliation but not sufficient for full state recovery on its own. This design preserves the Cloaking Protocol's concealment guarantee: even if the Jasper 2.0 identity vault were to be accessed by an adversary, the continuity snapshot reveals no information about the concealed operational state.

During concealment, while the system's true state S is hidden in the SCL shadow address space and the null surface is presented to external observers, Jasper 2.0 maintains a minimal identity proxy in the observable environment. The identity proxy is a stub component that responds to identity queries from other Seed Computing components — components that may need to verify that the cloaked system is still present and operational — with responses derived from the continuity snapshot. The proxy answers queries about the system's behavioral signature, its relational adjacency, and its operational status, returning responses that are consistent with the pre-concealment identity without exposing any information about the concealed state or the concealment event itself. The proxy's responses are generated by Jasper 2.0's identity inference engine, which derives plausible identity-consistent responses from the continuity snapshot using the same seed-based derivation operators that underpin the Seed Computing paradigm.

In the post-recovery phase, after the Deterministic Recovery Process has completed and the reconstructed state S' has been reintegrated into Aurora Runtime, Jasper 2.0 performs identity reconciliation. Reconciliation resolves any identity drift that

accumulated during the concealment interval — changes in the system's relational adjacency (other components may have formed or dissolved relationships during concealment), updates to the accumulated invariant history (the Invariant Engine may have processed additional invariant events during the null surface phase), and any divergence between the recovered state's behavioral signature and the signature recorded in the continuity snapshot. Reconciliation produces an updated identity record that incorporates both the recovered state's identity-critical information and the continuity snapshot's record of events during concealment, yielding a coherent, unbroken identity thread across the full cloaking cycle.

Jasper 2.0's role as a fallback for the Deterministic Recovery Process is significant and deserves careful characterization. In the failure mode where the Invariant Engine's seed bundle is irreparably incomplete and seed reconstruction from substrate signals has failed, the Deterministic Recovery Process cannot proceed through Stage 1. In this scenario, Jasper 2.0's continuity snapshot becomes the recovery source of last resort. The snapshot's seed bundle digest is used to attempt partial seed reconstruction; if sufficient seeds can be reconstructed from the digest and the identity vault's invariant history, recovery may proceed with a partially reconstructed seed bundle, accepting a bounded non-zero error $\varepsilon > 0$ in the recovered state. This fallback path is explicitly not the nominal recovery path and is documented in the system's operational specifications as a degraded-mode recovery with accompanying validation constraints.

Perhaps the most architecturally significant property of Jasper 2.0 in this context is its enabling of multi-cycle cloaking. A system that undergoes repeated concealment-recovery cycles — cloaked, recovered, recloaked — accumulates an identity history that would ordinarily fragment across cycles, as each recovery produces a state that is strictly speaking a reconstruction of a past state rather than the continuously evolved state that would have existed without concealment. Jasper 2.0 addresses this by maintaining the identity thread as a first-class object that evolves continuously across cycles, incorporating the identity-critical information from each recovery into a coherent longitudinal identity record. From the perspective of other Seed Computing components, a system that has been cloaked and recovered ten times is

indistinguishable from one that has operated continuously — because Jasper 2.0 has maintained the identity continuity that makes them equivalent.

SECTION 8

8. Applications

The Cloaking Protocol's combination of substrate-level concealment, deterministic recovery, and identity continuity preservation opens a range of applications within and adjacent to the Seed Computing paradigm. Five application areas are highlighted below.

- **Adversarial Resilience.** Systems operating in adversarial environments — environments in which external actors may attempt to extract operational state through forced inspection, side-channel analysis, or substrate-level probing — can leverage the Cloaking Protocol to resist these attacks. By triggering concealment in response to detected probing signals, the system presents a null surface that satisfies the inspection demand while the true operational state remains hidden. The adversary receives a plausible-looking operational facade rather than the actual state, and the system recovers full operation as soon as the inspection window closes. This application is particularly relevant for Seed Computing systems operating in zero-trust environments where substrate access cannot be fully controlled.
- **Zero-Downtime Substrate Migration.** The Teleportation Engine's concealment-aware migration protocol enables a cloaked system to be migrated across substrate boundaries without exposing sensitive operational state during transit. By initiating concealment before the migration begins, the system's true state is hidden in the SCL shadow address space, and only the concealment manifest and the null surface migrate across the network boundary. At the receiving substrate, the SCL reconstructs the shadow

address space from the manifest, and the system recovers full operation deterministically. The migration window — during which the system is most vulnerable to interception — therefore exposes only the null surface and the encrypted concealment manifest rather than the full operational state.

- **Fault Isolation and Recovery.** In cascading failure scenarios, where a single subsystem failure threatens to propagate through the broader Seed Computing environment, the Cloaking Protocol can isolate the degraded subsystem by cloaking it. Concealment severs the subsystem's observable connections to the rest of the environment — its execution context is hidden, its inter-component communication state is zeroed, and the null surface presents a quiescent facade — preventing failure propagation while the Invariant Engine prepares a clean recovery. When the underlying fault condition has been resolved, the Deterministic Recovery Process restores the subsystem to its pre-failure state, and it rejoins the environment through the standard recovery reintegration pathway.
- **Continuity Under Forced Inspection.** In regulatory or operational environments where systems are subject to mandatory audit or inspection, the Cloaking Protocol enables a compliant response that does not require disclosure of sensitive operational state. The system presents a null surface — a legally and operationally compliant operational facade — during the inspection window, while the true state is concealed. Upon completion of the inspection, the system recovers full operation deterministically. This application requires careful specification of the null surface template to ensure that it satisfies the inspection's compliance criteria; the Cloaking Protocol's null surface library supports customizable template sets for this purpose.
- **Seed Computing Research Infrastructure.** For researchers working within the Seed Computing paradigm, the Cloaking Protocol provides a controlled experimental sandbox capability. A researcher can trigger concealment of a running system state before executing a potentially destabilizing experiment — a novel invariant transformation, a seed

manipulation test, or a Topological Shift Model edge-case exploration — and then deterministically recover the original state if the experiment produces an undesirable outcome. This use of the Cloaking Protocol as an experimental safety net reduces the risk of irreversible state corruption in research environments and enables a more aggressive experimental methodology than would otherwise be prudent.

SECTION 9

9. Conclusion

This paper has presented the Cloaking Protocol: a formally grounded architectural mechanism for substrate-level state concealment and deterministic recovery within the Seed Computing paradigm. The protocol's three interlocking components — the Topological Shift Model, the Emergency State Hiding Model, and the Deterministic Recovery Process — constitute a coherent architectural layer that addresses a class of requirements for which classical fault-tolerance literature provides no precedent. Classical approaches assume observability as a prerequisite for recovery; the Cloaking Protocol demonstrates that this assumption is not fundamental but merely conventional, and that a recovery mechanism anchored in topological invariants can operate correctly in the complete absence of observable state.

The Unified Field Equation is the mathematical spine that makes this demonstration rigorous rather than heuristic. By expressing the recovery of the original state as a deterministic function of the seed bundle, the encoding key, the decoding operator, and the inverse topological shift, the UFE elevates recovery from a probabilistic best-effort operation to a provable deterministic computation — subject to the verifiable preconditions of seed bundle completeness and shift parameter availability. The failure mode analysis confirms that these preconditions are not merely assumed but actively enforced and defended by the Invariant Engine and, as a last resort, by Jasper 2.0's continuity checkpoint infrastructure.

The deep integration of the Cloaking Protocol with Aurora Runtime and Jasper 2.0 is what elevates it from an isolated mechanism to a full architectural layer. Aurora Runtime's three interface points — the Concealment Notification Interface, the Substrate Concealment Layer API, and the Recovery Handshake Protocol — provide the substrate-level cooperation required for concealment and recovery operations to be executed atomically with respect to the observable execution environment. Jasper 2.0's identity vault and continuity snapshot infrastructure extend the protocol's operational envelope to multi-cycle cloaking scenarios, maintaining identity continuity across arbitrarily many concealment-recovery cycles.

Three directions for future work are identified. First, formal proofs of recovery completeness: the current paper establishes the UFE's recovery semantics and characterizes convergence conditions, but formal machine-verified proofs of recovery completeness and bounded-error guarantees across the full range of topological shift types and seed bundle configurations remain as open problems. Second, hardware-level substrate concealment primitives: the SCL's shadow address space management is currently implemented in software, with performance implications at high CDI levels. Hardware-level primitives — specifically, memory management unit extensions that natively support shadow region exclusion — would substantially reduce concealment and recovery latency at deep CDI levels. Third, extension of the Topological Shift Model to multi-agent Seed Computing environments: the current model treats the system whose state is being concealed as a single computational entity. In multi-agent environments, where several Seed Computing components may undergo coordinated cloaking, the interactions between their respective topological shifts and seed bundles introduce a richer and more complex set of architectural questions that the current model does not address.

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